NPTEL Online Certification Courses
Indian Institute of Technology Kharagpur

Introduction to
Internet of Things
Assignment-Week 11

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

According to the lecture, which feature distinguishes a smart grid from a traditional electrical grid?

a. Energy distribution is only one directional.

b. Monitoring and restoration are always performed manually.

c. It enables bidirectional flows of energy with two-way communication and control.

d. Energy generation must occur only at centralized power plants.

Correct Answer: c. It enables bidirectional flows of energy with two-way communication and control.

Detailed Solution: The lecture describes a smart grid as a modernized grid that supports bidirectional energy flow and two-way communication and control, unlike the traditional grid's one-directional operation.

See Lecture 51 (Smart Grid - Part I, **slide 3**)

QUESTION 2:

What is the role of a smart meter in a smart home?

a. It generates electricity from renewable energy sources.

b. It provides the Smart Grid interface between the consumer and the energy service provider.

c. It replaces the home energy management system.

d. It operates only as a one-way energy meter without information exchange.

Correct Answer: b. It provides the Smart Grid interface between the consumer and the energy service provider.

Detailed Solution: Smart meters operate digitally and provide the interface for automated information exchange between the consumer side and the energy service provider.

See Lecture 51 (Smart Grid - Part I, **slide 14**)

QUESTION 3:

Under Time-of-Use pricing, when are consumers encouraged to operate smart appliances to reduce their energy cost?

a. During off-peak hours.

b. Only during on-peak hours.

c. Only when the grid has a blackout.

d. At a fixed time independent of the grid load.

Correct Answer: a. During off-peak hours.

Detailed Solution: Time-of-Use pricing encourages consumers to shift energy consumption to off-peak periods, when grid load is lower and the price of energy is lower.

See Lecture 51 (Smart Grid - Part I, **slide 20**)

QUESTION 4:

In net metering, the consumer's final bill depends on which two energy flows?

a. Reactive power and communication traffic.

b. Only the energy generated at centralized power plants.

c. Only the energy stored in the consumer's battery.

d. Energy inflow from the grid and energy outflow from the consumer to the grid.

Correct Answer: d. Energy inflow from the grid and energy outflow from the consumer to the grid.

Detailed Solution: The lecture states that net-metering bills depend on both the energy received by the consumer from the grid and the excess energy supplied by the consumer back to the grid.

See Lecture 51 (Smart Grid - Part I, **slide 21**)

QUESTION 5:

Which device is used in a smart grid operation center to sample voltage and current for monitoring the transmission system?

a. Phasor Measurement Unit (PMU).

b. Data Aggregator Unit (DAU).

c. Smart appliance controller.

d. Universal Power Line Bus.

Correct Answer: a. Phasor Measurement Unit (PMU).

Detailed Solution: A PMU samples voltage and current at its installed location. Multiple PMUs form a phasor network whose information can be analyzed by a centralized SCADA system.

See Lecture 52 (Smart Grid - Part II, **slide 3**)

QUESTION 6:

Which of the following is a function of distribution intelligence in the smart grid?

a. Preventing all use of smart meters.

b. Identifying the source of a power outage and supporting automatic power-flow restoration.

c. Restricting the grid to one-way communication.

d. Removing protection and control from feeder lines.

Correct Answer: b. Identifying the source of a power outage and supporting automatic power-flow restoration.

Detailed Solution: Distribution intelligence uses smart IoT devices and smart meters to identify outage sources, support automated switching, optimize real and reactive power, and protect feeder lines.

See Lecture 52 (Smart Grid - Part II, **slide 5**)

QUESTION 7:

In smart grid communication, what does a gateway connect?

a. Closely located smart meters to Data Aggregator Units (DAUs).

b. Only power plants directly to home appliances.

c. PMUs directly to plug-in electric vehicles.

d. Renewable generators directly to SCADA without intermediate communication.

Correct Answer: a. Closely located smart meters to Data Aggregator Units (DAUs).

Detailed Solution: The lecture explains that a gateway connects a group of closely located smart meters and acts as the communication link between those smart meters and the DAUs.

See Lecture 52 (Smart Grid - Part II, **slide 11**)

QUESTION 8:

Which type of attack compromises the availability of smart grid components or information by making them inaccessible when needed?

a. Net metering attack.

b. Time synchronization attack.

c. Data visualization attack.

d. Denial of Service (DoS) attack.

Correct Answer: d. Denial of Service (DoS) attack.

Detailed Solution: The lecture identifies attacks that compromise availability as Denial of Service attacks. Such attacks may block key signals and affect grid stability and observability.

See Lecture 52 (Smart Grid - Part II, **slide 17**)

QUESTION 9:

How is the Industrial Internet of Things (IIoT) described in the lecture?

a. An application of IoT in industries that links automation systems with enterprise, planning, and product lifecycle.

b. A consumer-only network for home entertainment devices.

c. A replacement for all industrial automation without networking.

d. A system used only for storing data without machine communication.

Correct Answer: a. An application of IoT in industries that links automation systems with enterprise, planning, and product lifecycle.

Detailed Solution: IIoT applies IoT concepts to industrial systems and connects automation with enterprise functions, planning, and product lifecycle activities.

See Lecture 53 (Industrial Internet of Things - Part I, **slide 3**)

QUESTION 10:

Which approach does the lecture associate with IIoT for adapting existing industrial systems?

a. Disconnect and isolate.

b. Rip and replace.

c. Wrap and re-use.

d. Remove and discard.

Correct Answer: c. Wrap and re-use.

Detailed Solution: The lecture states that IIoT is based on a 'wrap and re-use' approach rather than a 'rip and replace' approach, allowing existing industrial assets to be integrated.

See Lecture 53 (Industrial Internet of Things - Part I, **slide 6**)

QUESTION 11:

According to the lecture, which technology is identified as the key enabling technology for vehicle-to-vehicle and vehicle-to-road communications in Intelligent Transportation Systems?

a. Universal Power Line Bus.

b. Hadoop Distributed File System (HDFS).

c. Supervisory Control and Data Acquisition (SCADA).

d. Dedicated Short-Range Communications (DSRC).

Correct Answer: d. Dedicated Short-Range Communications (DSRC).

Detailed Solution: The lecture identifies DSRC as the key enabling technology for V2V and V2R communications in connected transportation systems.

See Lecture 54 (Industrial Internet of Things - Part II, **slide 5**)

QUESTION 12:

In IIoT-based mining applications, which environmental parameters can be monitored using wireless sensor networks?

a. Air quality, poisonous gases, and oxygen level inside mines.

b. Only the billing amount of mine workers.

c. Only the color of mining equipment.

d. Only the internet browsing history of operators.

Correct Answer: a. Air quality, poisonous gases, and oxygen level inside mines.

Detailed Solution: For mining safety, the lecture describes the use of RFID, Wi-Fi, and other wireless technologies for early warning, air-quality monitoring, poisonous-gas detection, and oxygen-level monitoring.

See Lecture 54 (Industrial Internet of Things - Part II, **slide 7**)

QUESTION 13:

Which Big Data characteristic refers to biases, noise, and abnormality in data?

a. Value.

b. Variability.

c. Veracity.

d. Volume.

Correct Answer: c. Veracity.

Detailed Solution: The lecture defines veracity in terms of bias, noise, and abnormality in data and emphasizes its importance for automated decision-making and machine learning.

See Lecture 55 (Data Handling and Analytics - Part I, **slide 9**)

QUESTION 14:

During data pre-processing, what is the purpose of redundancy mitigation?

a. To prevent data from being collected.

b. To intentionally duplicate every data item.

c. To convert all structured data into images.

d. To eliminate repeated data through detection, filtering, and compression.

Correct Answer: d. To eliminate repeated data through detection, filtering, and compression.

Detailed Solution: Redundancy mitigation removes repeated data using detection, filtering, and compression, helping avoid unnecessary transmission and storage.

See Lecture 55 (Data Handling and Analytics - Part I, **slide 18**)

QUESTION 15:

In Hadoop Distributed File System (HDFS), which node stores the actual data blocks?

a. Datanode.

b. Namenode.

c. Job Tracker.

d. YARN Resource Negotiator only.

Correct Answer: a. Datanode.

Detailed Solution: The Namenode maintains filesystem metadata, while Datanodes store the actual data. HDFS divides files into blocks and replicates those blocks across Datanodes.

See Lecture 55 (Data Handling and Analytics - Part I, **slide 23**)

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